

CPS I — Exercise Sheet 7

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Exercise 1: Linear-Time Properties

- Formalize P_i as a set of traces using set comprehension

- $P_1\{A_0A_1 \dots \in (2^{AP})^\omega \mid \forall i \in \mathbb{N}. a \in A_i \vee b \in A_i\}$
- $P_2\{A_0A_1 \dots \in (2^{AP})^\omega \mid (\forall i \in \mathbb{N}. (b \in A_i \implies \exists j \leq i. a \in A_j))\}$
- $P_3\{A_0A_1 \dots \in (2^{AP})^\omega \mid \forall i \in \mathbb{N}. (a \in A_i \implies \exists j > i. b \in A_j)\}$
- $P_4\{A_0A_1 \dots \in (2^{AP})^\omega \mid |\{i \in \mathbb{N} \mid a \in A_i\}| = 3\}$
- $P_5\{A_0A_1 \dots \in (2^{AP})^\omega \mid (|\{i \in \mathbb{N} \mid a \in A_i\}| = \infty) \rightarrow (|\{i \in \mathbb{N} \mid b \in A_i\}| = \infty)\}$
- $P_6\{A_0A_1 \dots \in (2^{AP})^\omega \mid |\{i \in \mathbb{N} \mid a \in A_i\}| < \infty\}$

- satisfies P_i

- $(\{a\}\{b\})^\omega$
- $\{a\}\{a, b\}^\omega$
- $(\{a\}\{b\})^\omega$
- $\{a\}\{a\}\{a\}\emptyset^\omega$
- $(\{a\}\{b\})^\omega$
- $\{a\}\emptyset^\omega$

- does not satisfy P_i

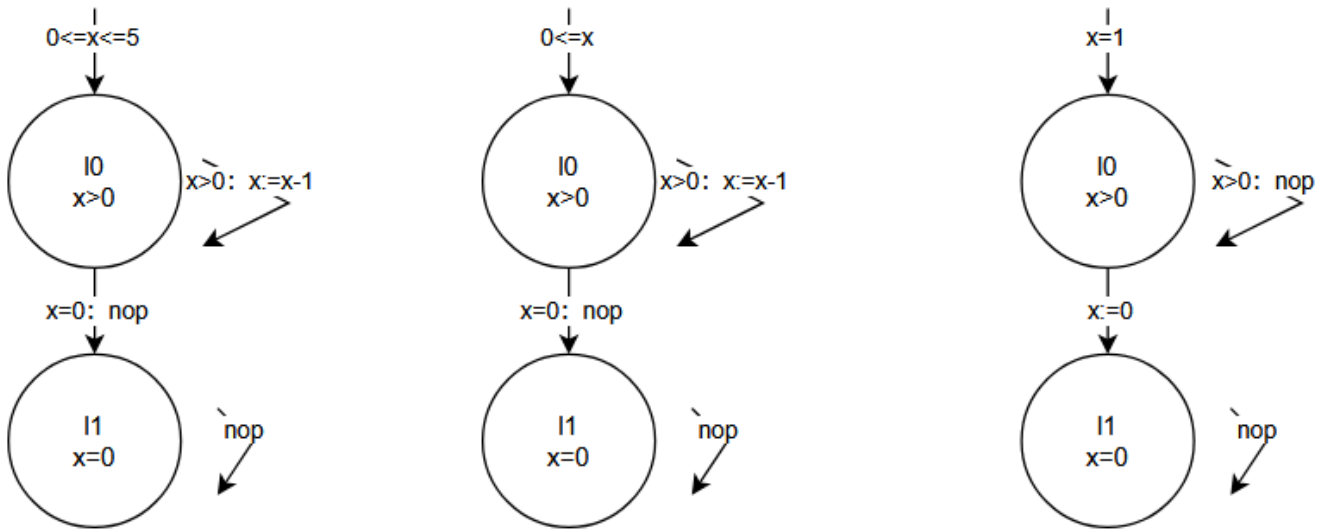
- $\emptyset(\{a\}\{b\})^\omega$
- $(\{b\}\{a\})^\omega$
- $\{a\}\emptyset^\omega$
- $\{b\}^\omega$
- $\emptyset\{a\}^\omega$
- $\{a\}^\omega$

- Explain whether or not the transition system below satisfies P

- false
- true
- true
- false
- true
- false

Exercise 2: Trace Inclusion

1.



2. First we find the Traces.

- $\text{Traces}(\mathcal{T}_{P_1}): \{\{x > 0\}^n \{x = 0\}^\omega \mid n \in \{0, 1, \dots, 5\}\}$
- $\text{Traces}(\mathcal{T}_{P_2}): \{\{x > 0\}^n \{x = 0\}^\omega \mid n \in \mathbb{N}_0\}$
- $\text{Traces}(\mathcal{T}_{P_{3a} \parallel P_{3b}}): \{\{x > 0\}^\omega\} \cup \{\{x > 0\}^k \{x = 0\}^\omega \mid k \in \mathbb{N}, k \neq 0\}$
- $\text{Traces}(\mathcal{T}_4): \{\{x > 0\}^\omega\} \cup \{\{x > 0\}^n \{x = 0\}^\omega \mid n \in \mathbb{N}_0\}$

1. $\mathcal{T}_{P_1} \subseteq \mathcal{T}_{P_2}$
2. $\mathcal{T}_{P_2} \not\subseteq \mathcal{T}_{P_1}$
3. $\mathcal{T}_{P_1} \not\subseteq \mathcal{T}_{P_{3a} \parallel P_{3b}}$
4. $\mathcal{T}_{P_{3a} \parallel P_{3b}} \not\subseteq \mathcal{T}_{P_1}$
5. $\mathcal{T}_{P_1} \subseteq \mathcal{T}_4$
6. $\mathcal{T}_4 \not\subseteq \mathcal{T}_{P_1}$
7. $\mathcal{T}_{P_2} \not\subseteq \mathcal{T}_{P_{3a} \parallel P_{3b}}$

8. $\mathcal{T}_{P_{3a} \parallel P_{3b}} \not\subseteq \mathcal{T}_{P_2}$
9. $\mathcal{T}_{P_2} \subseteq \mathcal{T}_4$
10. $\mathcal{T}_4 \not\subseteq \mathcal{T}_{P_2}$
11. $\mathcal{T}_{P_{3a} \parallel P_{3b}} \subseteq \mathcal{T}_4$
12. $\mathcal{T}_4 \not\subseteq \mathcal{T}_{P_{3a} \parallel P_{3b}}$

3. Eventually x=0

$$E = \{A_0 A_1 \dots \in (2^{\text{AP}})^\omega \mid \exists^\infty i \in \mathbb{N}. x = 0 \in A_i\}$$

As i mentioned the Traces in $(2)\mathcal{T}_{P_1} \models E\mathcal{T}_{P_2} \models E$

- $\mathcal{T}_{P_{3a} \parallel P_{3b}} \not\models E \text{ Exist Trace}\{x > 0\}^\omega$
- $\mathcal{T}_4 \not\models E \text{ Exist Trace}\{x > 0\}^\omega$